

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Electrical contact cleaner, Part Number 3824510
- Fuel System Clean Care Kit, Part Number 4919073

Additional Service Items

- Soft, clean, lint-free cloth

Initial Check

Use the following procedure for details on injector drain and leakage measurements. Refer to Procedure 005-236 in Section 5. (</qs3/pubsys2/xml/en/procedures/377/377-005-236.html>)

Preparatory Steps

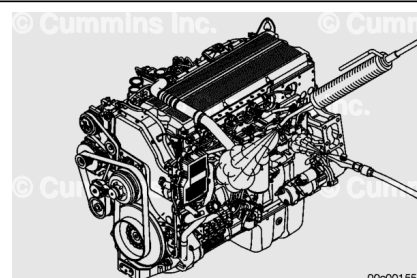
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠



Clean all fittings before disassembly. Dirt or contaminants can damage the fuel system.

Before servicing **any** fuel system components, (such as fuel lines, fuel pump, injectors, etc.), which would expose the fuel system or internal engine component to potential contaminants prior to disassembly, clean the fittings, mounting hardware, and the area around the component to be removed. Dirt or contaminants can be introduced into the fuel system and engine if the surrounding areas are **not** cleaned, resulting in damage to the fuel system and engine. Refer to Procedure 000-009 in Section 0.

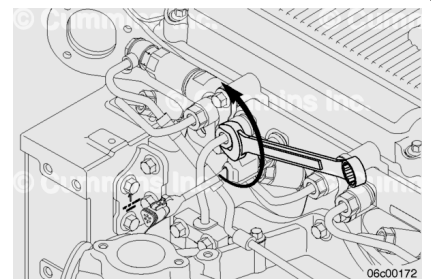
(/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)

Clean the injector supply line connections and mating components with electrical contact cleaner, Part Number 3824510, or equivalent.

To prevent damage from debris and contamination, cover, cap, or plug any openings as soon as possible when servicing the fuel system. Caps and plugs can be found in Fuel System Clean Care Kit, Part Number 4919073.

⚠ WARNING ⚠

Normal engine operation creates highly pressurized fuel in the fuel line which will remain in the fuel line after engine shutdown. Never open the fuel system when the engine is operating. Before servicing the fuel system, always loosen the pump-to-rail fuel line at the rail to vent the pressure. Keep hands clear of the line when loosening. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.



⚠ WARNING ⚠

When servicing the engine do not use the starting motor to rotate engine with a high-pressure fuel system joint open. Rotating the engine can create highly pressurized fuel in the fuel system. High-pressure fuel spray can penetrate the skin, resulting in serious personal injury or death.

Before servicing the fuel system, loosen the pump-to-rail line at the rail to vent the fuel pressure.

Note : A machined slot in this fitting directs the fuel spray toward the cylinder block.

Keep hands clear of the line when loosening the fuel rail nut.

Tighten the fuel rail nut.

Torque Value: 47 n•m [35 ft-lb]

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Remove the exhaust gas recirculation (EGR) crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Disconnect the internal actuator harness. Refer to Procedure 019-063 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-063.html>)
- Remove the high-pressure fuel line. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-051.html>)
- Remove the fuel connector retaining nut and fuel connector. Refer to Procedure 006-052 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-052.html>)

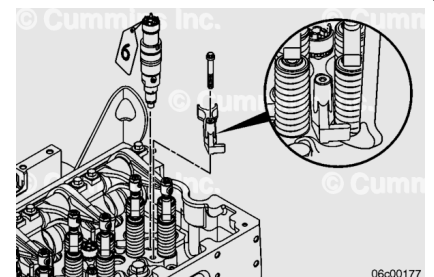
Remove

Note : If possible, provide a work area that will **not** have flying debris or dust. Gently place the removed injector(s) on a clean work table or container that has been cleaned of debris and oils. Immediately cover each injector to prevent contamination. Cover the engine if it will be left unattended to prevent contamination.

Note : Neighboring work areas may inadvertently contaminate the work area with debris from compressed air, air tools, or cleaning.

Remove the injector hold-down clamp capscrew.

Remove the injector hold-down clamp and injector together.



If necessary, use a small heel-bar to pry up on the injector clamp hold-down flange (part of the injector body just above the cylinder head casting). Verify that the fuel connector(s) has been removed prior to using a heel-bar to pry up on the injector(s).

Reused injectors **must** be installed with the same cylinder each time the injector is removed. Mark or tag the fuel injector with the cylinder number to keep track of the correct position of each fuel injector.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Do not use any abrasives (such as glass beading, sand paper, emery cloth, Scotch-Brite™ pads, etc) or metallic items (including wire brushes made of any metallic material) to clean the injectors. The use of any cleaning method other than safety solvent and a soft, clean, lint-free cloth will damage the nozzle holes and cause performance issues.

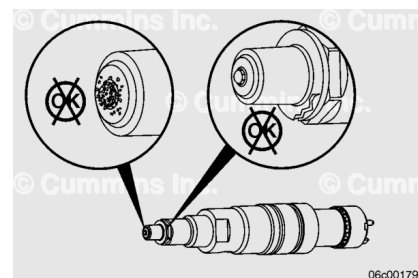
Clean the injector tip and body with electrical contact cleaner, Part Number 3824510, or equivalent, and a soft, clean, lint-free cloth.

Look for burrs on the inlet to the injector.

Check the nozzle holes for any signs of damage, such as hole erosion or hole plugging.

Spray electrical contact cleaner, Part Number 3824510, or equivalent, on the injector body. Inspect the high-pressure fuel connector interface for surface damage or discoloration, caused by excessive heat, which can allow high-pressure fuel to leak to the injector drain passage.

Inspect the horizontal sealing surface of the nozzle combustion shield for damage or signs of leaking.



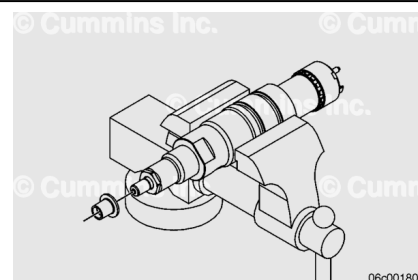
Disassemble

Note : The nozzle combustion shield must not be removed unless damaged.

Place the injector in a brass vise.

The nozzle combustion shield is pressed onto the tip of the injector.

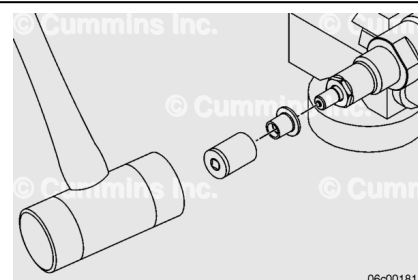
Grip the nozzle combustion shield at the injector tip end with pliers and twist to remove the shield from the tip of the injector. Use care **not** to damage the injector tip during removal.



Assemble

Place the injector in a brass vise.

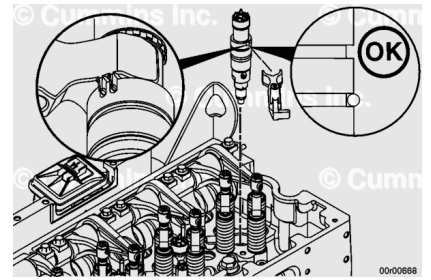
Place a 12 mm [0.47 in] deep well socket, or equivalent, over the nozzle combustion shield and tap the socket with a rubber hammer until the shield is flush with the shoulder of the injector.



Install

⚠ CAUTION ⚠

Avoid pushing on the injector terminals while installing or seating the injector. The injector terminals will be damaged if excessive force is applied.



Install new o-rings.

Inspect the injector bore for carbon deposits, if found, clean with clean lint free cloth at the end of clean 3/8 in [9.5 mm], or equivalent, dowel.

Lubricate the injector o-ring with clean engine oil.

Align the notches in the injector with the tab on the injector hold-down clamp.

Place the injector and hold-down clamp in the cylinder head in the proper orientation (fuel inlet toward the high-pressure fuel connector).

Push down on the injector hold-down clamp near the injector to make sure the injector is properly seated.

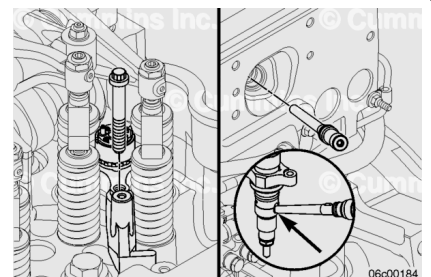
Use the injector shipping sleeve to make sure the injector is seated in the injector bore.

Note : If a new injector is being installed, a new fuel connector must also be used.

Start the injector hold-down clamp capscrew, but do **not** tighten.

To prevent damage to the fuel connector, push downward on the threaded end of the fuel connector to lift the fuel connector tip and help the connector tip seat properly against the injector.

Install the high-pressure fuel connector, making sure the end of the high-pressure fuel connector is in the injector inlet port.



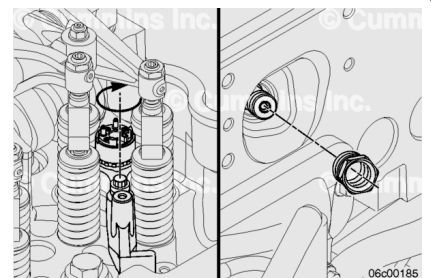
Tighten the injector hold-down clamp capscrew. (Step 1 of 2)

Torque Value: 20 n•m [177 in-lb]

Note : This is not the final torque for the injector hold-down clamp.

Lubricate the threads on the fuel connector retaining nut with clean engine oil.

Start the high-pressure fuel connector retaining nut and tighten partially. (Step 1 of 2)



Torque Value: 20 n•m [177 in-lb]

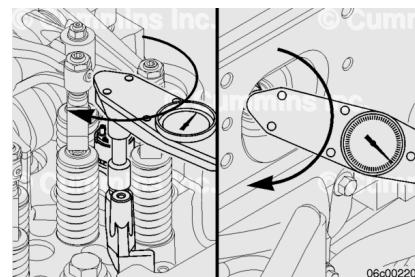
Note : This is **not** the final torque for the high-pressure fuel connector retaining nut.

Tighten the injector hold-down clamp capscrew. (Step 2 of 2)

Torque Value: 80 n•m [59 ft-lb]

Tighten the high-pressure fuel connector retaining nut. (Step 2 of 2)

Torque Value: 77 n•m [57 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Do not overtighten the injector harness. The injector terminals will be damaged if excessively overtightened.

- Connect the internal actuator harness to the injector. Refer to Procedure 019-063 in Section 19. (</qs3/pubsys2/xml/en/procedures/377/377-019-063.html>)
- Orient the injector wires so they will **not** interfere with a rocker lever or engine brake housing. If the rocker lever is able to come into contact with the injector harness, it will rub through the wire insulation and cause injector circuit fault codes.
- Install the high-pressure fuel lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/377/377-006-051.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/377/377-003-011-tr.html>)
- Install the EGR crossover tube. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/377/377-011-070-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.

Note : INSITE™ electronic service tool version 7.5.3 or newer is required to complete the injector reset.

Once a new injector is installed, an injector reset **must** be completed with INSITE™ electronic service tool.

Injector Reset Instructions

1. Connect INSITE™ electronic service tool.
2. Select "Advance ECM Data".
3. Select "Fuel Injector Reset".
4. Select each injector that needs to be reset and click "Reset".

Note : If multiple new injectors are installed, multiple injectors can be reset at the same time.

